EXPERIMENT -2

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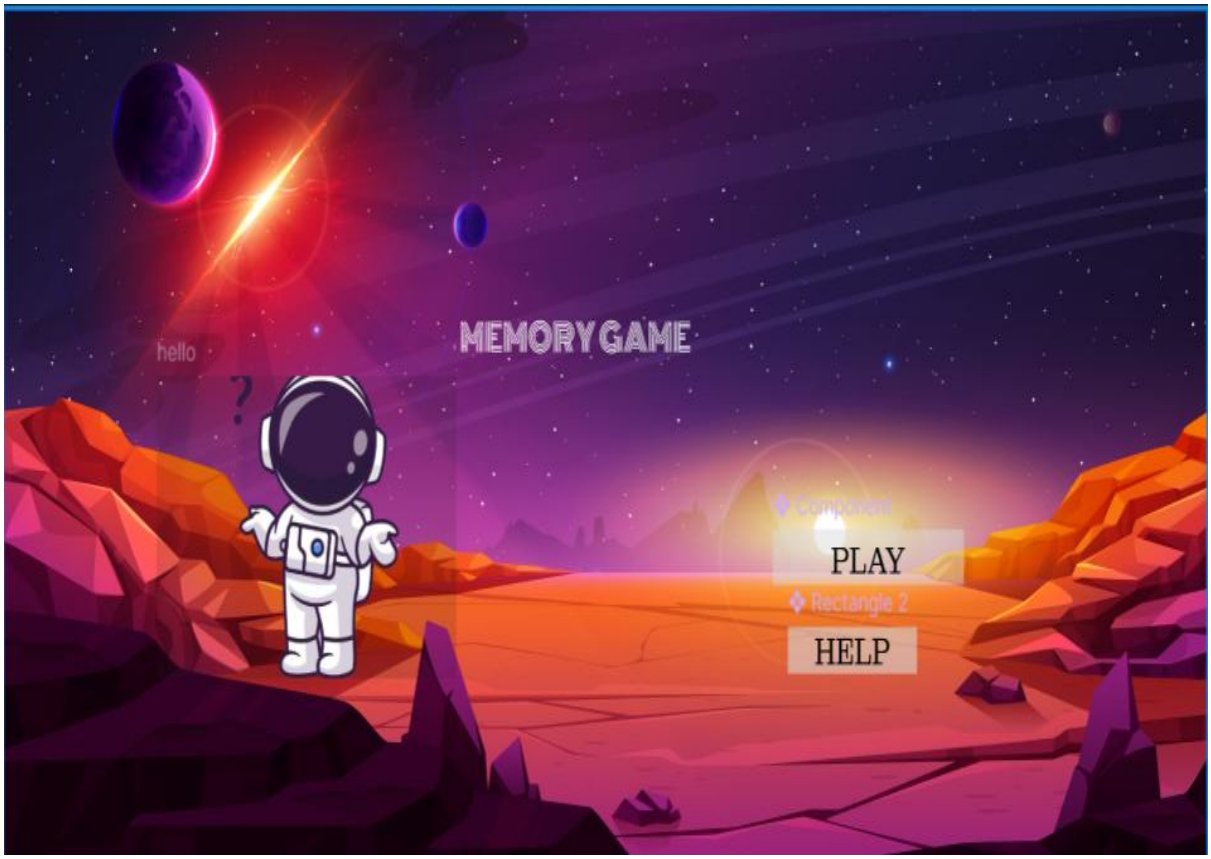
Design a UI where users recall visual elements (e.g, icons or text chunks). Evaluate the effect chunking memory

FRAME 1:

HOME / START SCREEN – MEMORY GAME



Screen Description

This is the **landing screen** of the Memory Game application. It introduces the user to the game in an engaging and visually immersive manner.

UI Elements

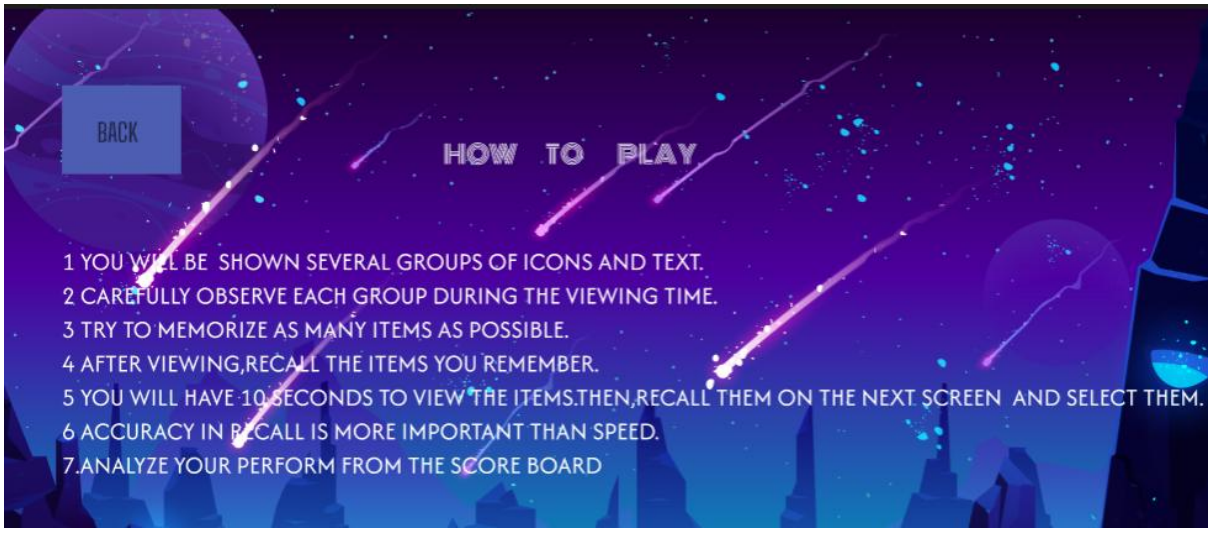
- **Title – “MEMORY GAME”**
  Positioned centrally to clearly communicate the purpose of the application.
- **Astronaut Character**
  Acts as a friendly visual guide, making the interface playful and inviting.
- **Buttons**
  - **PLAY** – Starts the memory recall task.
  - **HELP** – Navigates to the instruction page.
- **Background**
  - Space-themed environment with planets and stars.
  - Creates a calm yet engaging atmosphere that reduces anxiety before the task.

UX & Cognitive Aspect

- Simple layout reduces cognitive load.
- Clear call-to-action buttons improve usability.
- Gamified visuals increase user motivation and attention.

FRAME 2:

INSTRUCTION / HELP SCREEN – HOW TO PLAY



Screen Description

This screen provides **step-by-step instructions** explaining how the memory game works.

UI Elements

- **Heading – "HOW TO PLAY"**
- **Numbered Instructions (1–7)**
  - Explains observation, memorization, recall, and scoring.
- **Back Button**
  - Allows easy navigation to the home screen.
- **Consistent Space-Themed Background**

Chunking Analysis

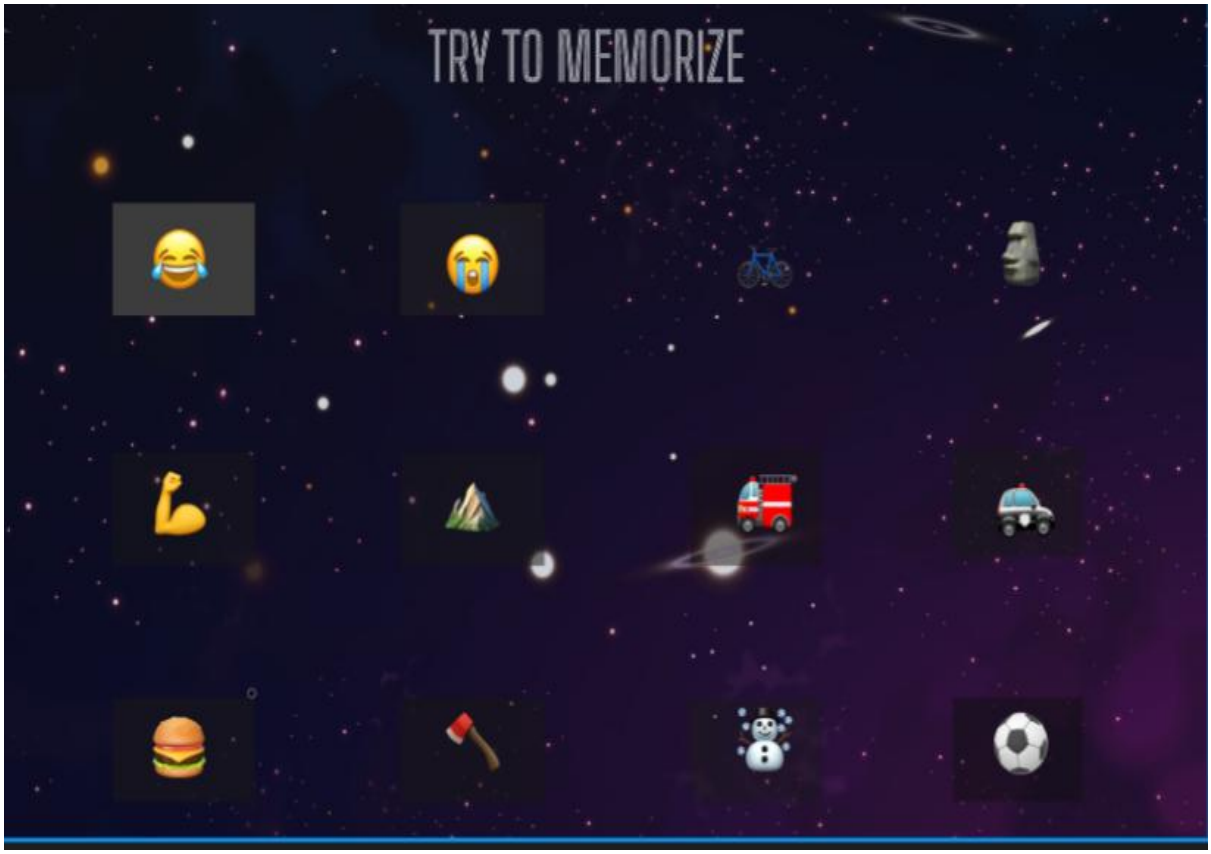
- Instructions are **broken into numbered steps**, which is a clear example of **textual chunking**.
- Each step contains **one idea**, making it easier to process and remember.
- Important rule highlighted: *Accuracy is more important than speed.*

Cognitive Benefit

- Reduces information overload.
- Improves comprehension and task readiness.
- Guides users smoothly into the memory task.

FRAME 3:

CHUNKING / MEMORIZATION PHASE – "TRY TO MEMORIZE"



Screen Description

This is the **visual encoding phase**, where users observe and memorize icons within a limited time.

UI Elements

- **Instruction Text – "TRY TO MEMORIZE"**
- **Grid of Icons**
    - Emojis and objects such as burger, axe, snowman, fire truck, football, etc.
    - Icons are evenly spaced and visually distinct.
- **Dark Space Background**
    - Helps icons stand out clearly.

Chunking Strategy

- Users can group items by:
    - **Category** (food, vehicles, objects, emojis)
    - **Visual similarity** (shape, color, size)

- Grid layout encourages **pattern recognition**.

Cognitive Effect

- Enhances short-term memory.
- Forces quick mental grouping due to time limitation.
- Promotes efficient visual scanning.

FRAME 4:

RECALL / SELECTION PHASE – “SELECT THE IMAGE YOU REMEMBER”

Screen Description

This screen tests the user’s **memory recall** by asking them to identify previously seen images.

UI Elements

- **Instruction Text – “SELECT THE IMAGE YOU REMEMBER”**

- **Multiple Icon Options**
    - Includes both correct items and distractors.
- **Selection Indicators**
    - Green dots indicate correct choices.
    - Red dots indicate incorrect selections.

Memory Evaluation

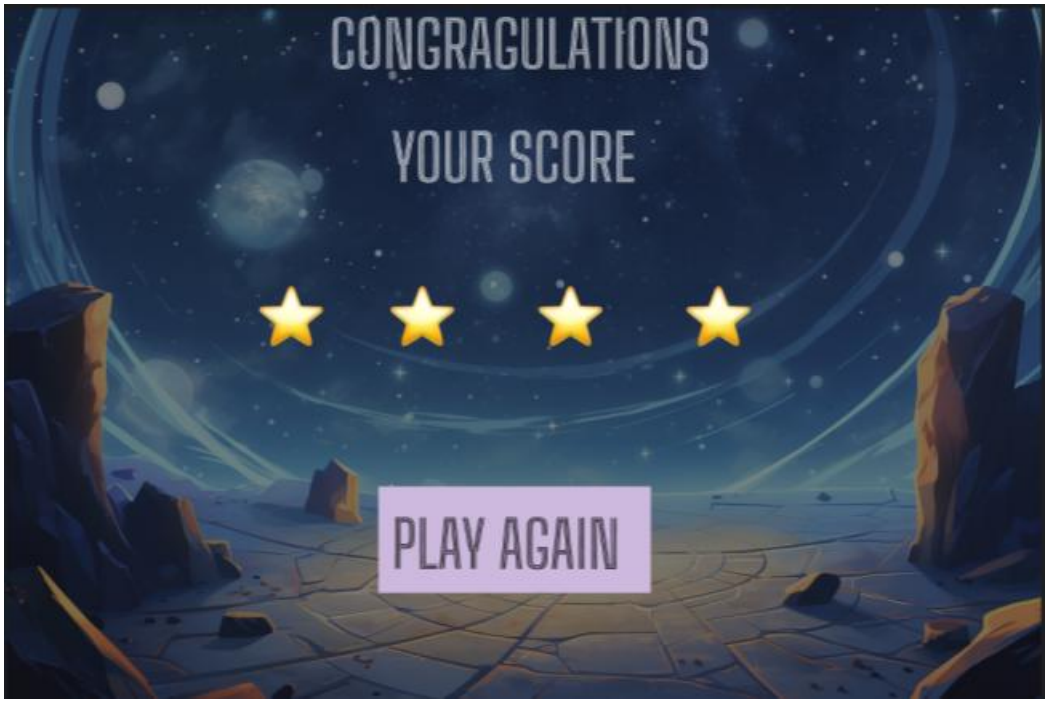
- Presence of distractors tests:
    - Accuracy of recall
    - Resistance to false memory
- Simple tap/click selection improves usability.

UX & Cognitive Benefit

- Clear feedback mechanism.
- Encourages careful recall rather than guessing.
- Reinforces learning through error recognition.

FRAME 5:

RESULT / SCORE SCREEN



Screen Description

This screen displays the **performance feedback** after completing the recall task.

UI Elements

- **Text – "CONGRATULATIONS – YOUR SCORE"**
- **Star Rating System**
  - Visual representation of performance.
- **"PLAY AGAIN" Button**
  - Allows repetition of the task for improvement.
- **Consistent Background**
  - Maintains immersion and continuity.

Feedback & Learning Aspect

- Immediate feedback helps users evaluate their memory performance.
- Star-based scoring is easy to understand and motivating.
- Replay option supports practice and improvement.

OVERALL UX & CHUNKING EVALUATION

- The UI effectively applies **chunking principles** by:
  - Breaking tasks into clear stages.
  - Using grids and categories for visual grouping.
  - Presenting instructions in structured steps.
- Gamified elements improve engagement and reduce stress.
- The design successfully evaluates the **effect of chunking on user memory recall**.

PROTOTYPE LINK:

https://www.figma.com/design/shS8EQhQnnIlfYBCkyDsVf/Untitled?node-id=0-1&m=dev&t=bPb1rq6p5RBtlz2D-1